

# MASK-INDUCED DISPLACEMENT IN AUDIO XAI VIA LOGIT TRAJECTORY DECOMPOSITION

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## ABSTRACT

Perturbation-based XAI methods for audio classifiers often estimate feature importance by masking spectrogram regions and crediting output changes to the retained signal. Yet they typically assume the fill (the mask replacement) is negligible. We propose logit-space trajectory decomposition to examine this assumption. An on-axis component captures output along a line connecting the fully filled (occluded) spectrogram to the fully retained original; an off-axis component captures perpendicular displacement. We evaluate across three fills, three audio classifiers, and 1,100 AudioSet clips. We demonstrate that no fill is acoustically neutral under full occlusion: ZERO activates silence and GAUSSIAN NOISE activates broadband noise. Under partial masking, 41–77% of output displacement is off-axis, with the direction of the residual stable across mask retention fraction and specific to each model–fill combination. Attribution methods are unevenly exposed to off-axis displacement through their sampling and weighting strategies, revealing apparatus-dependence. Where the off-axis residual is stable and low-dimensional, its structure affords mitigation.

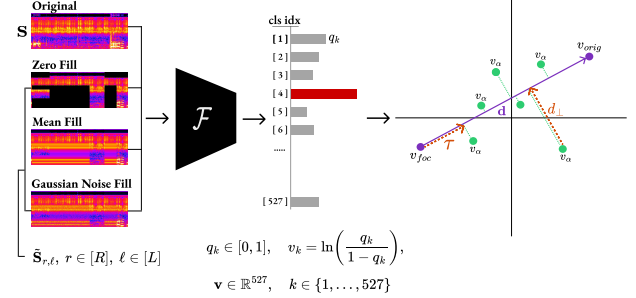
**Index Terms**— explainable AI, feature attribution, audio classification, perturbation-based explanation, spectrogram masking.

## 1. INTRODUCTION

Perturbation-based eXplainable AI (XAI) methods explain neural classifier predictions by masking input features and crediting output changes to the retained signal. However, masking cannot remove input features, only substitute them with a replacement signal, referred to herein as *fill*; every output reflects the classifier’s joint response to both, introducing missingness bias [1] and risking out-of-distribution model responses [2]. We ask: *how predictably does model output change given the mixture of retained and fill signal, and what structure underlies any such departure?*

The natural approach to evaluating fill choice is ablation: test various fills, measure attribution faithfulness, and select the most faithful. However, without ground-truth explanations against which to validate, evaluation methods have been shown to be circular: faithfulness evaluation rewards alignment between the fill used in attribution and evaluation, hyperparameters can produce any desired ranking [3], and models may infer class from mask shape during pixel-flipping [4], causing persistent feature attribution disagreement [5, 6]. Conclusions drawn from perturbation-based XAI are apparatus-dependent: what is measured reflects methodological choices as much as model behaviour. We therefore forgo faithfulness evaluation in favour of a geometric investigation of model output under partial masking.

We measure model output holistically in logit space, establishing a reference axis from the fully occluded condition (FOC,  $\alpha = 0$ ) to the original output ( $\alpha = 1$ ) as a geometric reference for decomposition. Displacement along this axis is on-axis; perpendicular



**Fig. 1.** Method overview. Four spectrograms (original and three fills) are passed through classifier  $\mathcal{F}$ , producing logit vectors  $\mathbf{v} \in \mathbb{R}^{527}$ . Each  $\mathbf{v}_{\alpha}$  denotes the logit vector at retention fraction  $\alpha$ . The logit-space trajectory is decomposed into a scalar on-axis component  $\tau$  (along the attribution axis  $\mathbf{d} = \mathbf{v}_{\text{orig}} - \mathbf{v}_{\text{foc}}$ ) and a scalar off-axis magnitude  $d_{\perp}$ , whose direction is captured by the residual vector  $\mathbf{v}_{\perp} \in \mathbb{R}^{527}$ .

displacement is off-axis. Logit trajectory decomposition characterises both components across three fills (ZERO, MEAN, GAUSSIAN NOISE), three classifiers (AST+SA, PANNS-SA, PANNS+SA), and 1,100 AudioSet clips. We make the following contributions:

1. **Partial Mask Analysis:** We propose logit-space trajectory decomposition, showing that 41–77% of partial-mask output displacement is off-axis and invisible to scalar attribution methods (Section 4.2). This displacement concentrates in stable low-dimensional subspaces specific to each model–fill combination, with consequences for how mask sampling and surrogate weighting interact with fill-specific geometry (Section 4.3).
2. **Fully Occluded Condition:** We show no fill constitutes an information-absent baseline, with each fill activating a distinct AudioSet class profile under full occlusion (Section 4.1).

All code and trained models are available at: [https://github.com/ne3k0/partial\\_mask\\_geometry\\_xai](https://github.com/ne3k0/partial_mask_geometry_xai)

## 2. RELATED WORK

Perturbation-based XAI methods, such as KernelSHAP [7], LIME [8], and RISE [9], have seen comparatively little study in the audio domain relative to vision [10], despite widespread audio classifier deployment [11]. Nonetheless, they have been applied across audio tasks, from singing voice detection [12] and music tagging [13] to automatic speech recognition [14], audio content analysis [15], and speech emotion recognition [16]. Perturbation methods have also been used to generate causal explanations [17], revealing model-specific feature reliance in audio classifiers [18], first noted by Sturm’s Clever Hans critique [19] and confirmed unresolved a decade

later [20]. Learned-surrogate approaches train decoder networks on internal classifier representations [21, 22], with faithfulness gains at the cost of surrogate interpretability.

Fill non-neutrality has been explored in feature masking literature [23], including work framed as feature removal [24], where fill dominates faithfulness rankings [25], and in integrated gradients, where baseline choice similarly shapes attribution [26]. Black-box treatment and interpretability motivate perturbation-based methods. Core differences between audio and visual modalities, namely that mel spectrograms encode non-isotropic frequency-temporal information and SpecAugment [27] applies zero-fill temporal and spectral masking during training, mean that approaches inherited from vision require dedicated study in the audio domain.

### 3. METHOD

#### 3.1. Models & Experiment Setup

We evaluate three mel spectrogram audio classifiers trained on AudioSet [28]: PANNS+SA [29], PANNS-SA [29], and AST+SA [30]. The two PANNS variants share the same training data, isolating SpecAugment as the sole training difference. AST+SA is evaluated with SpecAugment only, as omitting it causes overfitting and notable performance degradation. Our perturbation pipeline replicates each model’s signal chain exactly.

The evaluation dataset is drawn from the AudioSet eval split. Fifty clips are sampled from 22 classes<sup>1</sup>, yielding 1,100 clips, applied to each model and fill condition. Classes were selected to span diverse acoustic signal types, not sampled at random from AudioSet. Model output is always measured against the eval clip’s ground truth (GT) label, not the model’s top-1 prediction.

#### 3.2. Spectrogram Segmentation, Sampling & Perturbation

**STFT-domain Perturbation:** Fill is applied in complex STFT space:

$$\tilde{\mathbf{Z}} = \mathbf{m} \odot \mathbf{Z} + (1 - \mathbf{m}) \odot \mathbf{Z}_{\text{fill}}, \quad (1)$$

where  $\odot$  denotes the Hadamard product,  $\mathbf{m} \in \{0, 1\}^{F \times T}$  is a binary retention mask, and  $\mathbf{Z}_{\text{fill}}$  the fill signal. Each model’s signal chain  $\Psi : \mathbb{C}^{F \times T} \rightarrow \mathbb{R}^{M \times T}$  maps  $\tilde{\mathbf{Z}}$  to the log mel spectrogram  $\tilde{\mathbf{S}}$  the classifier receives, discarding phase information.

**Segmentation and Sampling:** Masks are defined over a coarser  $M \times X$  cell grid, with time boundaries at multiples of  $T/10$  (each cell  $\approx 100$  ms), then upsampled to STFT resolution: frequency bins via mel filterbank support, time frames via precomputed hard assignment to their enclosing coarse window, yielding a binary mask  $\mathbf{m} \in \{0, 1\}^{F \times T}$ . The grid is partitioned into  $K = K_F \times K_T = 25$  contiguous rectangular blocks with boundaries redrawn each seed, following the coalition-sampling strategy of RISE [9], where each block constitutes a coalition of cells sampled together. With  $R = 300$  seeds and  $L = 32$  samples per seed, each clip yields 9,600 partial masks;  $\alpha \in [0, 1]$  is the fraction of cells retained per partial mask, binomially distributed around 0.5 (Figure 2).

We adopt RISE-style sampling because redrawing boundaries each seed ensures any cell appears across many coalitions, avoiding structural assumptions about co-occurring regions, and, over sufficiently many samples and configurations, minimising positional bias from cells differing in how strongly they drive the model output.

With fixed grids, cell membership within each block is permanent, so output variation at a given  $\alpha$  partly reflects which blocks happen to contain important cells rather than  $\alpha$  alone, limiting any test of LIME and KernelSHAP’s assumption that  $\alpha$  proxies proximity to the original signal. Masks are deterministic, enabling direct comparison of fill effects across conditions.

Let  $Z_{f,t}$  be the element of  $\mathbf{Z}$  at bin  $f$ , frame  $t$ , with magnitude  $|Z_{f,t}|$  and phase  $\theta_{f,t}$ ;  $\mu_f = \frac{1}{T} \sum_{t=1}^T |Z_{f,t}|$  is the time-averaged magnitude for each frequency bin. The fill elements  $Z_{\text{fill},f,t}$  are defined by one of three strategies:

1. **ZERO:**  $Z_{\text{fill},f,t} = 0$
2. **MEAN:**  $Z_{\text{fill},f,t} = \mu_f e^{j\theta_{f,t}}$  — replaces magnitude with the time-averaged  $\mu_f$ .
3. **GAUSSIAN NOISE:**  $Z_{\text{fill},f,t} = |\epsilon_{f,t}| e^{j\theta_{f,t}}$ , where  $\epsilon_{f,t} \sim \mathcal{N}(0, \mu_f^2)$  — magnitude is half-normal, scaled per frequency bin by the clip’s average spectral magnitude  $\mu_f$ . Drawn once per clip and held constant across all masks.

#### 3.3. Logit-Space Trajectory Analysis

Model outputs are analysed in logit space:  $v_k = \ln(q_k/(1 - q_k))$  for each class  $k$ , giving  $\mathbf{v} \in \mathbb{R}^{527}$  (one dimension per AudioSet class).  $\bar{q}$  denotes mean sigmoid confidence across clips. This linearises the sigmoid outputs for geometric analysis. Let  $\mathbf{v}_{\text{orig}}$  and  $\mathbf{v}_{\text{foc}}$  denote the logit vectors of a clip’s original and fully occluded inputs respectively, and let  $\mathbf{d} = \mathbf{v}_{\text{orig}} - \mathbf{v}_{\text{foc}} \in \mathbb{R}^{527}$  be the attribution axis (hereafter  $\mathbf{d}$ ; Fig. 1). Although  $\mathbf{d}$  is clip-specific, the quantities derived from it are normalised, preserving cross-clip comparability. For a masked output at retention fraction  $\alpha$ , let  $\Delta\mathbf{v} = \mathbf{v}_\alpha - \mathbf{v}_{\text{foc}} \in \mathbb{R}^{527}$  denote its displacement from the fully occluded output. We decompose  $\Delta\mathbf{v}$  along  $\mathbf{d}$ :

$$\tau = \frac{\Delta\mathbf{v} \cdot \mathbf{d}}{\|\mathbf{d}\|^2}, \quad d_\perp = \|\Delta\mathbf{v} - \tau \mathbf{d}\|, \quad d_{\text{total}} = \|\Delta\mathbf{v}\|. \quad (2)$$

$\tau \in \mathbb{R}$  is the scalar projection onto  $\mathbf{d}$  (so  $\tau \mathbf{d}$  in the second term is a scaled vector), anchored so that  $\tau = 0$  at FOC and  $\tau = 1$  at the original output.  $d_\perp$  is the magnitude of the displacement orthogonal to  $\mathbf{d}$ . The additive surrogate assumption requires  $\tau \approx \alpha$ . Large  $d_\perp$  evidences output activation in directions the attribution axis cannot account for. The scalar  $\tau_{\text{gt}} = \Delta v_{\text{gt}}/d_{\text{gt}}$  gives the normalised ground-truth (GT) class logit displacement by which attribution methods assign feature importance to each partial mask, where  $d_{\text{gt}}$  is the GT-class component of  $\mathbf{d}$ . We report  $\bar{\sigma}_\tau$  and  $\bar{\sigma}_{d_\perp}$ , the mean per-clip standard deviations of  $\tau$  and  $d_\perp$ , and  $d_\perp/d_{\text{total}}$ , the mean off-axis fraction of total output displacement.

#### 3.4. Structure of the Off-Axis Residual

The residual vector  $\mathbf{v}_\perp = \Delta\mathbf{v} - \tau \mathbf{d} \in \mathbb{R}^{527}$  is perpendicular to  $\mathbf{d}$  by construction. To characterise the structure of  $\mathbf{v}_\perp$  within each retention bin, we estimate its within-bin covariance. Fix a mel spectrogram classifier  $\mathcal{F}$  and fill  $\mathbf{Z}_{\text{fill}}$ . For each retention bin  $b$  (corresponding to  $\alpha \in [\alpha_b, \alpha_{b+1})$ , with  $B = 20$  bins), pool all  $\mathbf{v}_\perp$  vectors across the set of 22 evaluation classes and all clips, yielding  $n_b$  vectors. Retention bins follow a binomial distribution, with minimum  $n_b = 9,891$  across valid bins, giving a sample-to-dimension ratio of at least 19:1. Pooling across clips is intentional: it enables detection of shared off-axis structure across the clip ensemble, not just within individual clips. The within-bin covariance is:

$$\Sigma_b = \mathbb{E}[\mathbf{v}_\perp \mathbf{v}_\perp^\top \mid \alpha \in b] - \mu_b \mu_b^\top \in \mathbb{R}^{527 \times 527}, \quad (3)$$

<sup>1</sup>Bagpipes, Boing, Chicken/rooster, Didgeidoo, Dog, Drum kit, Frog, Frying food, Gunshot/gunfire, Hair dryer, Harmonica, Heart sounds/heartbeat, Insect, Owl, Rain, Rub, Sewing machine, Speech, Thunder, Timpani, Train, Whispering.

**Table 1.** Top-2 predicted classes per fill at FOC ( $N = 1,100$  clips, 22 classes).  $\bar{q}$ : mean sigmoid confidence.  $\sigma$  omitted for ZERO (signal-independent).

| Model    | Fill           | Top-1 class | $\bar{q}$ | $\sigma$ | $n$  | Top-2 class | $\bar{q}$ | $\sigma$ | $n$  |
|----------|----------------|-------------|-----------|----------|------|-------------|-----------|----------|------|
| AST+SA   | ZERO           | Silence     | .608      | –        | 1100 | Music       | .269      | –        | 1100 |
|          | MEAN           | Mains hum   | .691      | .220     | 857  | Hum         | .598      | .236     | 864  |
|          | GAUSSIAN NOISE | White noise | .466      | .145     | 969  | Noise       | .176      | .092     | 350  |
| PANNS-SA | ZERO           | Silence     | .327      | –        | 1100 | Music       | .186      | –        | 1100 |
|          | MEAN           | Buzzer      | .216      | .123     | 273  | Hum         | .173      | .129     | 178  |
|          | GAUSSIAN NOISE | White noise | .342      | .136     | 618  | Static      | .234      | .121     | 545  |
| PANNS+SA | ZERO           | Music       | .187      | –        | 1100 | Silence     | .035      | –        | 1100 |
|          | MEAN           | Sine wave   | .512      | .241     | 295  | Siren       | .226      | .164     | 120  |
|          | GAUSSIAN NOISE | Static      | .229      | .104     | 348  | White noise | .159      | .068     | 401  |

where  $\mu_b$  is the bin mean vector. Subtracting it centres  $\Sigma_b$ ; since each  $\mathbf{v}_\perp$  is perpendicular to its clip-specific  $\mathbf{d}$ ,  $\mu_b$  averages across different perpendicular subspaces and carries no consistent geometric interpretation. All estimates use a numerically stable single-pass batch update [31].

Eigendecomposition of  $\Sigma_b$  gives eigenvalues  $\lambda_1 \geq \dots \geq \lambda_{526} \geq 0$ . We summarise the spectral concentration of each bin with the participation ratio:

$$\text{PR}_b = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}, \quad (4)$$

which gives the effective number of dimensions occupied by  $\mathbf{v}_\perp$  in bin  $b$ .  $\text{PR}_b = 1$  for a rank-1 distribution concentrated along a single direction;  $\text{PR}_b = 526$  in the isotropic limit, since  $\mathbf{v}_\perp$  is constrained orthogonal to  $\mathbf{d}$  by construction. Table 3 reports the mean PR across valid bins.

To test whether the dominant off-axis direction is consistent across retention fraction, we extract the leading eigenvector  $\mathbf{e}_b$  of  $\Sigma_b$  for each bin and compute the pairwise absolute cosine similarity  $c_{b,b'} = |\mathbf{e}_b \cdot \mathbf{e}_{b'}|$  across all bin pairs, with absolute value handling eigenvector sign ambiguity. The mean and minimum pairwise cosine similarity, denoted  $\bar{c}$  and  $\min c_{b,b'}$ , characterise directional stability across  $\alpha$ . We additionally report  $\bar{\rho}$ , the mean over bins of the fraction of off-axis variance explained by the leading direction (Table 3).

## 4. RESULTS & DISCUSSION

### 4.1. Fill Non-Neutrality Under Full Occlusion

An information-absent baseline requires  $\mathbf{v}_{\text{foc}} = \mathbf{0}$ , that is,  $q_k = 0.5$  for all  $k$ . No fill can guarantee this for a discriminatively trained classifier, and any fill carries an acoustic character to which trained models respond. ZERO fill is signal-independent, yielding identical output distributions across all clips and classes within each model.  $\mathbf{v}_{\text{foc}}$  therefore has zero clip-to-clip variance and  $\sigma$  is not reported for ZERO fill. MEAN and GAUSSIAN NOISE fills are signal-dependent, introducing clip-level variance absent under ZERO fill (Table 1).

The top-1 class under ZERO fill differs notably across architectures. AST+SA and PANNS-SA both assign Silence ( $\bar{q} = 0.608$  and  $\bar{q} = 0.327$  respectively), congruent with zero fill’s acoustic character, whereas PANNS+SA assigns Music ( $\bar{q} = 0.187$ ).

MEAN fill is the most differentiated condition across architectures. AST+SA shows strong top-1 consensus (Mains hum, 857/1100,  $\bar{q} = 0.691$ ). PANNS-SA shows weak top-1 consensus (273/1100,  $\bar{q} = 0.216$ ), distributing probability mass across a large set of classes. PANNS+SA concentrates on tonal-periodic classes (Sine wave, Siren)

**Table 2.** Modal top-1 class and mean  $\bar{q}$  per source class under MEAN at FOC. Colours indicate semantic group: <sup>†</sup>electrical interference (Mains hum, Hum, Buzzer), <sup>§</sup>tonal periodic (Sine wave, Siren).

|             | AST+SA                       | PANNS-SA                  | PANNS+SA                     |
|-------------|------------------------------|---------------------------|------------------------------|
| Frying food | Mains hum <sup>†</sup> – .83 | Hum <sup>†</sup> – .07    | Siren <sup>§</sup> – .13     |
| Drum kit    | Mains hum <sup>†</sup> – .78 | Buzzer <sup>†</sup> – .06 | Sound fx – .18               |
| Owl         | Sine wave <sup>§</sup> – .39 | Hum <sup>†</sup> – .09    | Sine wave <sup>§</sup> – .49 |
| Heartbeat   | Sine wave <sup>§</sup> – .37 | Music – .12               | Sine wave <sup>§</sup> – .47 |
| Whispering  | Mains hum <sup>†</sup> – .24 | Hum <sup>†</sup> – .09    | Sine wave <sup>§</sup> – .41 |
| Bagpipes    | Mains hum <sup>†</sup> – .14 | Buzzer <sup>†</sup> – .17 | Siren <sup>§</sup> – .14     |

(Table 2). GAUSSIAN NOISE fill yields the most semantically consistent responses across models. All three converge on broadband noise classes, with AST+SA reaching the highest consensus (969/1100: White noise,  $\bar{q} = 0.466$ ) (Table 1).

MEAN and GAUSSIAN NOISE fills inherit spectral properties from the source clip, making fill-intrinsic discriminativeness and source signal structure inseparable at the clip level, a confound that propagates into partial-mask attribution at all retention fractions.

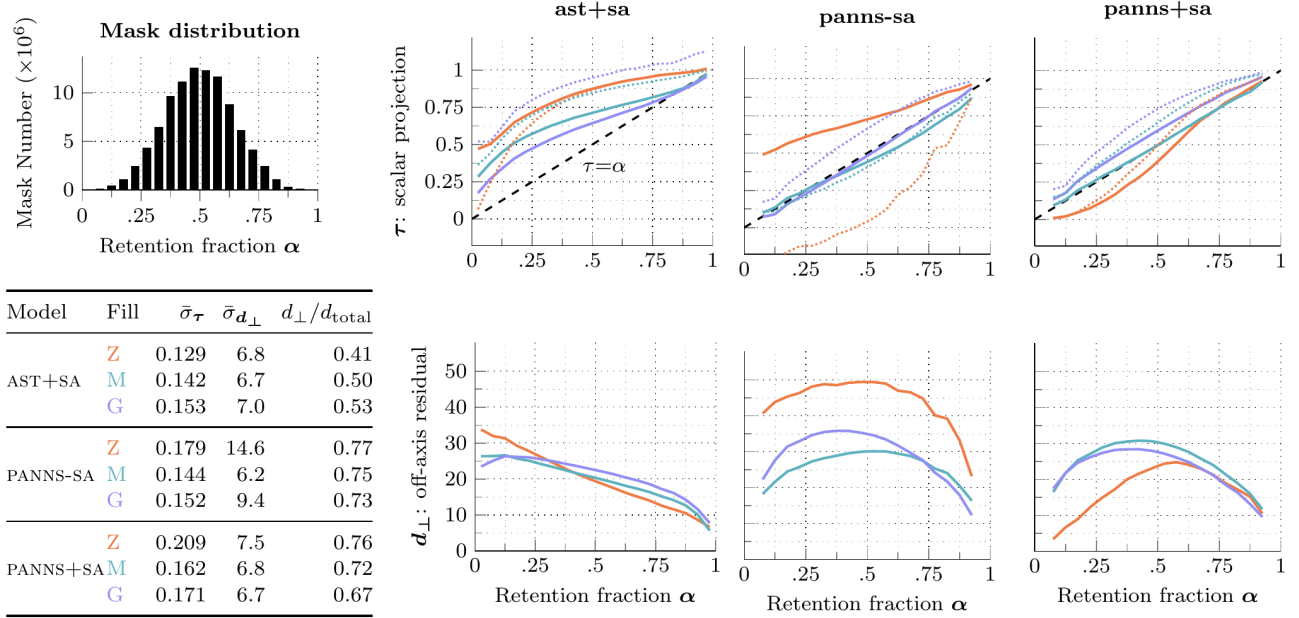
### 4.2. Logit-Space Trajectories Under Partial Masking

Decomposing the output trajectory geometrically from the FOC reveals that between 41% and 77% of total output displacement is orthogonal to the attribution axis  $\mathbf{d}$ , captured by  $d_\perp/d_{\text{total}}$  (Fig. 2). This displacement is invisible to scalar target-class methods by construction: LIME and KernelSHAP model GT-class output as a linear function of retained features and RISE tracks the scalar GT-class output, weighting all masks equally regardless of  $\alpha$ . Under high  $d_\perp$ , attribution weights are assigned in the presence of fill-driven model responses the surrogate cannot observe.

AST+SA is the least affected ( $d_\perp/d_{\text{total}} \in [0.41, 0.53]$ ), with  $d_\perp$  generally decreasing as  $\alpha$  increases. LIME’s proximity kernel, which weights masks by similarity to the original input, favours high- $\alpha$  coalitions, coinciding with the region of lowest  $d_\perp$  for AST+SA and reducing surrogate exposure to unaccounted-for off-axis activation. Under GAUSSIAN NOISE and MEAN fills, both signal-dependent,  $\tau_{\text{gt}}$  starts at approximately 0.52 and 0.36 at  $\alpha = 0.025$  respectively, with Gaussian fill  $\tau_{\text{gt}} > 1$  at  $\alpha > 0.75$ .

Both PANNs architectures show substantially larger off-axis displacement than AST+SA ( $d_\perp/d_{\text{total}} \in [0.67, 0.77]$ ). SpecAugment reduces clip-level instability under ZERO fill ( $\bar{\sigma}_\perp$ : 14.6  $\rightarrow$  7.5) and dramatically suppresses zero-fill  $d_\perp$  at low retention fractions, consistent with learned invariance to zero-fill masking during training. MEAN and GAUSSIAN NOISE fills show minimal change across the two PANNs variants. For PANNS-SA, Gaussian and mean fills

**Fig. 2.** Logit-space trajectory analysis across 95.04M partial masks (22 classes, 50 clips, 9,600 masks/clip). Fill: **Z** = zero, **M** = mean, **G** = Gaussian. Row 1: mask distribution and mean on-axis projection  $\tau$  by  $\alpha$ . Solid line = full logit vector; dotted = GT class ( $\tau_{\text{gt}}$ ); dashed = ideal ( $\tau = \alpha$ ). Row 2: per-fill summary table with  $\bar{\sigma}_\tau$ ,  $\bar{\sigma}_{d_\perp}$ ,  $d_\perp/d_{\text{total}}$ ; mean off-axis residual  $d_\perp$  by  $\alpha$ .



track the  $\tau = \alpha$  reference well yet  $d_\perp$  remains substantial. On-axis predictability and off-axis contamination coexist.

Both PANNS variants exhibit an inverted-U  $d_\perp$  profile, peaking around mid- $\alpha$  where fill and retained signal are most evenly mixed. Bernoulli block sampling with  $p=0.5$  concentrates the majority of masks in this region by construction, precisely where off-axis displacement is highest. KernelSHAP’s Shapley kernel assigns lower regression weights to mid- $\alpha$  coalitions when fitting the surrogate, coinciding with the peak of the PANNS  $d_\perp$  profile. It is therefore better positioned to reduce the impact of PANNS off-axis behaviour. Fill shapes the logit-space trajectory in directions invisible to scalar attribution methods, with architecture and fill-specific consequences that  $\tau$  alone cannot reveal.

### 4.3. Structure of the Off-Axis Component

PANNS-SA under zero fill exhibits the most structured off-axis residual (Table 3): PR = 8.4 indicates that  $\mathbf{v}_\perp$  occupies fewer than nine effective dimensions.  $\bar{c} = 0.95$  confirms the leading direction is consistent across  $\alpha$ , and 32.1% of off-axis variance is concentrated in a single eigenvector. A stable, low-dimensional off-axis profile of this kind is the most tractable target for future correction strategies.

SpecAugment training reduces this stability markedly. PANNS+SA falls below  $\bar{c} \leq 0.82$  across all three fills, with  $\min c_{b,b'}$  as low as 0.02 under GAUSSIAN NOISE, indicating that the leading off-axis direction shifts substantially across retention fraction. Direction instability across the mask ensemble makes post-hoc correction strategies less tractable.

AST+SA presents a distinct profile:  $\bar{c}$  remains high across all three fills (0.87–0.95), with  $\min c_{b,b'} \geq 0.74$  for zero and mean, indicating that the leading off-axis direction is stable across  $\alpha$ . However,  $\text{PR} \in [13.4, 19.1]$  and  $\bar{\rho} \in [18.5, 23.5]\%$  show that variance is distributed across more dimensions than PANNS-SA under ZERO. AST+SA is directionally consistent but structurally diffuse.

**Table 3.** Off-axis residual structure per model–fill condition. PR: participation ratio (effective dimensionality).  $\bar{c}$ : mean pairwise eigenvector cosine similarity.  $\min c_{b,b'}$ : worst-case similarity.  $\bar{\rho}$ : variance fraction explained by leading eigenvector. Fill: Z = ZERO, M = MEAN, G = GAUSSIAN NOISE.

| Model    | Fill | PR    | $\bar{c}$ | $\min c_{b,b'}$ | $\bar{\rho}$ (%) |
|----------|------|-------|-----------|-----------------|------------------|
| AST+SA   | Z    | 19.14 | .95       | .74             | 18.5             |
|          | M    | 13.37 | .94       | .76             | 23.5             |
|          | G    | 15.13 | .87       | .25             | 20.0             |
| PANNS-SA | Z    | 8.39  | .95       | .68             | 32.1             |
|          | M    | 19.56 | .86       | .20             | 15.3             |
|          | G    | 16.03 | .90       | .52             | 18.5             |
| PANNS+SA | Z    | 17.62 | .71       | .24             | 21.6             |
|          | M    | 16.75 | .82       | .28             | 16.7             |
|          | G    | 21.20 | .61       | .02             | 13.1             |

## 5. CONCLUSION

Masking fill and attribution method are not neutral experimental choices in perturbation-based XAI. Across three architectures and 1,100 AudioSet clips, none of the three tested fills constitutes an information-absent baseline, with each activating a distinct class profile under full occlusion. Under partial masking, between 41 and 77% of output displacement is orthogonal to the attribution axis. Logit-space geometric decomposition provides a principled black-box diagnostic for how fill and attribution method shape model outputs. For certain model–fill combinations, the off-axis component concentrates in a stable, low-dimensional subspace, affording mitigation opportunities that an isotropic residual would not. Evaluating any such correction, however, requires faithfulness metrics that themselves depend on fill choice. Understanding the causal mechanics behind the fill-specific output geometry requires opening the black box to examine intermediate activations.

## 6. COMPLIANCE WITH ETHICAL STANDARDS

This is a computational study using the publicly available AudioSet dataset [28]. No human subjects were involved and no ethical approval was required.

## 7. ACKNOWLEDGMENTS

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